

1 Install Docker

Step 1

Update system packages.

```
sudo apt update
```

Step 2

Install Docker.

```
sudo apt install docker.io -y
```

Step 3

Start Docker service.

```
sudo systemctl start docker  
sudo systemctl enable docker
```

Step 4

Add your user to Docker group.

```
sudo usermod -aG docker $USER  
newgrp docker
```

Step 5

Verify Docker installation.

```
docker --version
```

Check Docker is working.

```
docker ps
```

2 Install kubectl

Step 6

Install **kubectl**.

```
sudo snap install kubectl --classic
```

Verify installation.

```
kubectl version -client
```

3 Install Minikube

Step 7

Download Minikube.

```
curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-linux-amd64
```

Step 8

Install Minikube.

```
sudo install minikube-linux-amd64 /usr/local/bin/minikube
```

Step 9

Check version.

```
minikube version
```

4 Start Kubernetes Cluster

Step 10

Start the cluster.

```
minikube start
```

Step 11

Verify cluster status.

```
kubectl get nodes
```

Expected output:

```
minikube    Ready    control-plane
```

5 Create Project Folder

Step 12

Create a working directory.

```
mkdir ai-kubernetes
cd ai-kubernetes
```

6 Create AI Application

Step 13

Create Python file.

```
nano app.py
```

Paste:

```
from flask import Flask, request, jsonify
app = Flask(__name__)
@app.route('/')
def home():
    return "AI Model Running Successfully"
@app.route('/predict', methods=['POST'])
def predict():
    data = request.json['number']
    result = data * 2
    return jsonify({'result': result})
if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

Save:

```
CTRL + X
Y
Enter
```

7 Create Dockerfile

Step 14

```
nano Dockerfile
```

Paste:

```
FROM python:3.10
WORKDIR /app
COPY . .
RUN pip install flask
EXPOSE 5000
CMD ["python", "app.py"]
```

Save the file.

8 Build Docker Image

Step 15

Configure Docker to use Minikube.

```
eval $(minikube docker-env)
```

Step 16

Build Docker image.

```
docker build -t ai-model .
```

Verify image.

```
docker images
```

9 Create Kubernetes Deployment

Step 17

Create deployment file.

```
nano deployment.yaml
```

Paste:

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ai-model-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: ai-model
  template:
    metadata:
      labels:
        app: ai-model
    spec:
      containers:
        - name: ai-container
          image: ai-model
          imagePullPolicy: Never
          ports:
            - containerPort: 5000
```

Save the file.

Step 18

Apply deployment.

```
kubectl apply -f deployment.yaml
```

Verify pods.

```
kubectl get pods
```

Create Service

Step 19

Create service file.

```
nano service.yaml
```

Paste:

```
apiVersion: v1
kind: Service
metadata:
  name: ai-model-service
spec:
  type: NodePort
  selector:
    app: ai-model
  ports:
    - port: 5000
      targetPort: 5000
      nodePort: 30007
```

Save.

Step 20

Apply service.

```
kubectl apply -f service.yaml
```

Check services.

```
kubectl get services
```

Access Application

Step 21

Run:

```
minikube service ai-model-service
```

Your browser will open and display:

AI Model Running Successfully

1 2 Scale Pods (Important Step)

Step 22

Increase number of pods.

```
kubectl scale deployment ai-model-deployment --replicas=5
```

Step 23

Verify scaling.

```
kubectl get pods
```

You should see **5 pods running**.

1 3 Open Kubernetes Dashboard

Step 24

Launch dashboard UI.

```
minikube dashboard
```

You can now see:

Pods
Deployments
Services
Nodes

Final Architecture

```
graph TD
    User --> K8S[Kubernetes Service]
    K8S --> DC[Deployment Controller]
    DC --> Pods[Pod1 Pod2 Pod3 Pod4 Pod5]
```

Final Terminal Workflow Summary

```
Install Docker
Install kubectl
Install Minikube
↓
Start Cluster
↓
Create AI App
↓
Create Docker Image
↓
Create Deployment
↓
Create Service
↓
Access Application
↓
Scale Pods
↓
Open Dashboard
```

✅ This is the **complete terminal-based Kubernetes practical**.